Deep Learning

Theoretical Exercises - Lesson 3 - Chapter 4

Exercises on the book "Deep Learning" written by Ian Goodfellow, Yoshua Bengio, and Aaron Courville.
Exercises and solutions by T. Méndez and G. Schuster

FS 2026

1 Exercises on Numerical Computation

1. Given is the loss function

$$f(\boldsymbol{x}) = \frac{1}{4}\,x_1^4 + x_1^3 - \frac{17}{4}\,x_1^2 - 6\,x_1 + \frac{1}{5}\,x_2^4 + \frac{6}{5}\,x_2^3 + 89. \tag{1.1}$$

   (a) Determine the gradient and the Hessian matrix of $f(\boldsymbol{x})$.

   (b) Determine all critical points and, if possible, also determine the type of the critical point (local minimum, local maximum or saddle point) using the Hessian matrix.

   (c) Determine the global minimum.

   (d) Search for a minimum by using the steepest descent method. In doing so, start at the following points and use the indicated learning rates:

      i. Start point: $\boldsymbol{x}^{(0)} = [0\ \ \text{-}0.5]^T$, learning rate: $\epsilon = 0.1$

      ii. Start point: $\boldsymbol{x}^{(0)} = [\text{-}2.5\ \ 4]^T$, learning rate: $\epsilon = 0.05$

   Finish the learning algorithm after 10 iterations and check whether you have found the global minimum.

   (e) Repeat Exercise (d) with Newton's method.

Solution:

   (a) To determine the gradient and the Hessian matrix, all first and second partial derivatives must be calculated:

$$\begin{aligned}
\frac{\partial f(\boldsymbol{x})}{\partial x_1} &= x_1^3 + 3\,x_1^2 - \frac{17}{2}\,x_1 - 6 \\
\frac{\partial f(\boldsymbol{x})}{\partial x_2} &= \frac{4}{5}\,x_2^3 + \frac{18}{5}\,x_2^2 \\
\frac{\partial^2 f(\boldsymbol{x})}{\partial x_1^2} &= 3\,x_1^2 + 6\,x_1 - \frac{17}{2} \\
\frac{\partial^2 f(\boldsymbol{x})}{\partial x_2^2} &= \frac{12}{5}\,x_2^2 + \frac{36}{5}\,x_2 \\
\frac{\partial^2 f(\boldsymbol{x})}{\partial x_1 \partial x_2} &= 0.
\end{aligned}$$

Thus, the gradient is

$$\nabla_{\boldsymbol{x}} f(\boldsymbol{x}) = \begin{bmatrix} \frac{\partial f(\boldsymbol{x})}{\partial x_1} \\ \frac{\partial f(\boldsymbol{x})}{\partial x_2} \end{bmatrix} = \begin{bmatrix} x_1^3 + 3\,x_1^2 - \frac{17}{2}\,x_1 - 6 \\ \frac{4}{5}\,x_2^3 + \frac{18}{5}\,x_2^2 \end{bmatrix}$$

and the Hessian matrix is

$$\boldsymbol{H}(f)(\boldsymbol{x}) = \begin{bmatrix} \frac{\partial^2 f(\boldsymbol{x})}{\partial x_1^2} & \frac{\partial^2 f(\boldsymbol{x})}{\partial x_1 \partial x_2} \\ \frac{\partial^2 f(\boldsymbol{x})}{\partial x_1 \partial x_2} & \frac{\partial^2 f(\boldsymbol{x})}{\partial x_2^2} \end{bmatrix} = \begin{bmatrix} 3\,x_1^2 + 6\,x_1 - \frac{17}{2} & 0 \\ 0 & \frac{12}{5}\,x_2^2 + \frac{36}{5}\,x_2 \end{bmatrix}. \tag{1.2}$$

(b) Critical points are defined as points where the gradient is the zero vector, thus

$$\nabla_{\boldsymbol{x}} f(\boldsymbol{x}) = \begin{bmatrix} x_1^3 + 3\,x_1^2 - \frac{17}{2}\,x_1 - 6 \\ \frac{4}{5}\,x_2^3 + \frac{18}{5}\,x_2^2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

or

$$x_1^3 + 3\,x_1^2 - \frac{17}{2}\,x_1 - 6 = 0 \tag{1.3}$$

$$\frac{4}{5}\,x_2^3 + \frac{18}{5}\,x_2^2 = 0, \tag{1.4}$$

respectively. The zeroes of equation (1.3) are

$$x_{1,a} = -4.572, \quad x_{1,b} = -0.603, \quad x_{1,c} = 2.175$$

and the zeroes of equation (1.4) are

$$x_{2,a} = 0, \quad x_{2,b} = -4.5,$$

resulting in the following critical points:

$$\boldsymbol{c}_1 = \begin{bmatrix} -4.572 \\ 0 \end{bmatrix}, \qquad \boldsymbol{c}_2 = \begin{bmatrix} -0.603 \\ 0 \end{bmatrix}, \qquad \boldsymbol{c}_3 = \begin{bmatrix} 2.175 \\ 0 \end{bmatrix},$$

$$\boldsymbol{c}_4 = \begin{bmatrix} -4.572 \\ -4.5 \end{bmatrix}, \qquad \boldsymbol{c}_5 = \begin{bmatrix} -0.603 \\ -4.5 \end{bmatrix}, \qquad \boldsymbol{c}_6 = \begin{bmatrix} 2.175 \\ -4.5 \end{bmatrix}.$$

To also determine the type, the Hessian matrix (1.2) has to be evaluated at every critical point $\boldsymbol{c}_i$. This results in

$$\boldsymbol{H}(f)(\boldsymbol{c}_1) = \begin{bmatrix} 26.7793 & 0 \\ 0 & 0 \end{bmatrix}, \qquad \boldsymbol{H}(f)(\boldsymbol{c}_2) = \begin{bmatrix} -11.0278 & 0 \\ 0 & 0 \end{bmatrix},$$
$$\boldsymbol{H}(f)(\boldsymbol{c}_3) = \begin{bmatrix} 18.7485 & 0 \\ 0 & 0 \end{bmatrix}, \qquad \boldsymbol{H}(f)(\boldsymbol{c}_4) = \begin{bmatrix} 26.7793 & 0 \\ 0 & 16.2 \end{bmatrix},$$
$$\boldsymbol{H}(f)(\boldsymbol{c}_5) = \begin{bmatrix} -11.0278 & 0 \\ 0 & 16.2 \end{bmatrix}, \qquad \boldsymbol{H}(f)(\boldsymbol{c}_6) = \begin{bmatrix} 18.7485 & 0 \\ 0 & 16.2 \end{bmatrix},$$

which means that $\boldsymbol{c}_4$ and $\boldsymbol{c}_6$ are local minima, since both eigenvalues of their Hessian matrices are positive, and $\boldsymbol{c}_5$ is a saddle point, since one eigenvalue of its Hessian matrices is positive and the other one is negative. For the critical points $\boldsymbol{c}_1$, $\boldsymbol{c}_2$ and $\boldsymbol{c}_3$ the second derivative test is inconclusive with respect to the second variable. Thus, $\boldsymbol{c}_1$ and $\boldsymbol{c}_3$ are either local minima or saddle points and $\boldsymbol{c}_2$ is either a local maximum or a saddle point. By analyzing higher-order derivations, one can see that all three points ($\boldsymbol{c}_1$, $\boldsymbol{c}_2$ and $\boldsymbol{c}_3$) are saddle points.

(c) To determine which critical point corresponds to the global minimum, the value of the loss function (1.1) can be calculated for every critical point. This results in the following values

$$f(\boldsymbol{c}_1) = 41.2599, \qquad f(\boldsymbol{c}_2) = 90.8865, \qquad f(\boldsymbol{c}_3) = 71.7287,$$
$$f(\boldsymbol{c}_4) = 13.9224, \qquad f(\boldsymbol{c}_5) = 63.5490, \qquad f(\boldsymbol{c}_6) = 44.3912,$$

which makes $\boldsymbol{c}_4$ the global minimum. Actually, it would be sufficient to calculate the loss function for the local minima $\boldsymbol{c}_4$ and $\boldsymbol{c}_6$, since we know that all other critical points are saddle points.

(d) The update rule for the steepest descent algorithm is

$$\begin{aligned} \boldsymbol{x}^{(i)} &= \boldsymbol{x}^{(i-1)} - \epsilon \cdot \nabla_{\boldsymbol{x}} f(\boldsymbol{x}^{(i-1)}) \\ &= \begin{bmatrix} x_1^{(i-1)} \\ x_2^{(i-1)} \end{bmatrix} - \epsilon \cdot \begin{bmatrix} \left(x_1^{(i-1)}\right)^3 + 3\left(x_1^{(i-1)}\right)^2 - \frac{17}{2} x_1^{(i-1)} - 6 \\ \frac{4}{5}\left(x_2^{(i-1)}\right)^3 + \frac{18}{5}\left(x_2^{(i-1)}\right)^2 \end{bmatrix}. \end{aligned}$$

Starting from point $\boldsymbol{x}^{(0)} = [0 \;\; \text{-}0.5]^T$, using a learning rate of $\epsilon = 0.1$ and performing 10 steps, results in the point $\boldsymbol{x}^{(10)} = [2.0957 \;\; \text{-}4.5316]^T$, which corresponds to the local minimum $\boldsymbol{c}_6$. The steps of this calculation are shown in the following table:

$\boldsymbol{x}^{(0)}$	$\boldsymbol{x}^{(1)}$	$\boldsymbol{x}^{(2)}$	$\boldsymbol{x}^{(3)}$	$\boldsymbol{x}^{(4)}$	$\boldsymbol{x}^{(5)}$
$\begin{bmatrix} 0.0000 \\ -0.5000 \end{bmatrix}$	$\begin{bmatrix} 0.6000 \\ -0.5800 \end{bmatrix}$	$\begin{bmatrix} 1.5804 \\ -0.6855 \end{bmatrix}$	$\begin{bmatrix} 2.3797 \\ -0.8289 \end{bmatrix}$	$\begin{bmatrix} 1.9559 \\ -1.0307 \end{bmatrix}$	$\begin{bmatrix} 2.3225 \\ -1.3255 \end{bmatrix}$
$\boldsymbol{x}^{(6)}$	$\boldsymbol{x}^{(7)}$	$\boldsymbol{x}^{(8)}$	$\boldsymbol{x}^{(9)}$	$\boldsymbol{x}^{(10)}$	
$\begin{bmatrix} 2.0257 \\ -1.7717 \end{bmatrix}$	$\begin{bmatrix} 2.2853 \\ -2.4568 \end{bmatrix}$	$\begin{bmatrix} 2.0675 \\ -3.4434 \end{bmatrix}$	$\begin{bmatrix} 2.2587 \\ -4.4457 \end{bmatrix}$	$\begin{bmatrix} 2.0957 \\ -4.5316 \end{bmatrix}$	

By starting from a different point $\boldsymbol{x}^{(0)} = [\text{-}2.5 \;\; 4]^T$, using a learning rate of $\epsilon = 0.05$

and again performing 10 steps, the point $\boldsymbol{x}^{(10)} = [\text{-}4.5721 \ \ \text{-}4.4885]^T$ is found, which corresponds to the global minimum $\boldsymbol{c}_4$. The steps of this second calculation are shown in the following table:

$\boldsymbol{x}^{(0)}$	$\boldsymbol{x}^{(1)}$	$\boldsymbol{x}^{(2)}$	$\boldsymbol{x}^{(3)}$	$\boldsymbol{x}^{(4)}$	$\boldsymbol{x}^{(5)}$
$\begin{bmatrix} -2.5000 \\ 4.0000 \end{bmatrix}$	$\begin{bmatrix} -3.4188 \\ -1.4400 \end{bmatrix}$	$\begin{bmatrix} -4.3270 \\ -1.6938 \end{bmatrix}$	$\begin{bmatrix} -4.6237 \\ -2.0158 \end{bmatrix}$	$\begin{bmatrix} -4.5531 \\ -2.4196 \end{bmatrix}$	$\begin{bmatrix} -4.5783 \\ -2.9068 \end{bmatrix}$
$\boldsymbol{x}^{(6)}$	$\boldsymbol{x}^{(7)}$	$\boldsymbol{x}^{(8)}$	$\boldsymbol{x}^{(9)}$	$\boldsymbol{x}^{(10)}$	
$\begin{bmatrix} -4.5700 \\ -3.4453 \end{bmatrix}$	$\begin{bmatrix} -4.5728 \\ -3.9461 \end{bmatrix}$	$\begin{bmatrix} -4.5718 \\ -4.2911 \end{bmatrix}$	$\begin{bmatrix} -4.5722 \\ -4.4450 \end{bmatrix}$	$\begin{bmatrix} -4.5721 \\ -4.4885 \end{bmatrix}$	

(e) The update rule for Newton's algorithm is

$$\boldsymbol{x}^{(i)} = \boldsymbol{x}^{(i-1)} - H(f)(\boldsymbol{x}^{(i-1)})^{-1}\nabla_{\boldsymbol{x}} f(\boldsymbol{x}^{(i-1)})$$

$$= \begin{bmatrix} x_1^{(i-1)} \\ x_2^{(i-1)} \end{bmatrix} - \begin{bmatrix} \frac{1}{3\left(x_1^{(i-1)}\right)^2 + 6\,x_1^{(i-1)} - \frac{17}{2}} & 0 \\ 0 & \frac{1}{\frac{12}{5}\left(x_2^{(i-1)}\right)^2 + \frac{36}{5}\,x_2^{(i-1)}} \end{bmatrix} \begin{bmatrix} \left(x_1^{(i-1)}\right)^3 + 3\left(x_1^{(i-1)}\right)^2 - \frac{17}{2}\,x_1^{(i-1)} - 6 \\ \frac{4}{5}\left(x_2^{(i-1)}\right)^3 + \frac{18}{5}\left(x_2^{(i-1)}\right)^2 \end{bmatrix}.$$

Starting from point $\boldsymbol{x}^{(0)} = [0 \ \ \text{-}0.5]^T$ and performing 10 steps, results in the point $\boldsymbol{x}^{(10)} = [\text{-}0.6033 \ \ \text{-}0.0004]^T$, which corresponds to the critical point $\boldsymbol{c}_2$ (a saddle point). The steps of this calculation are shown in the following table:

$\boldsymbol{x}^{(0)}$	$\boldsymbol{x}^{(1)}$	$\boldsymbol{x}^{(2)}$	$\boldsymbol{x}^{(3)}$	$\boldsymbol{x}^{(4)}$	$\boldsymbol{x}^{(5)}$
$\begin{bmatrix} 0.0000 \\ -0.5000 \end{bmatrix}$	$\begin{bmatrix} -0.7059 \\ -0.2333 \end{bmatrix}$	$\begin{bmatrix} -0.6042 \\ -0.1134 \end{bmatrix}$	$\begin{bmatrix} -0.6033 \\ -0.0560 \end{bmatrix}$	$\begin{bmatrix} -0.6033 \\ -0.0278 \end{bmatrix}$	$\begin{bmatrix} -0.6033 \\ -0.0139 \end{bmatrix}$
$\boldsymbol{x}^{(6)}$	$\boldsymbol{x}^{(7)}$	$\boldsymbol{x}^{(8)}$	$\boldsymbol{x}^{(9)}$	$\boldsymbol{x}^{(10)}$	
$\begin{bmatrix} -0.6033 \\ -0.0069 \end{bmatrix}$	$\begin{bmatrix} -0.6033 \\ -0.0035 \end{bmatrix}$	$\begin{bmatrix} -0.6033 \\ -0.0017 \end{bmatrix}$	$\begin{bmatrix} -0.6033 \\ -0.0009 \end{bmatrix}$	$\begin{bmatrix} -0.6033 \\ -0.0004 \end{bmatrix}$	

Starting from a different point $\boldsymbol{x}^{(0)} = [\text{-}2.5 \ \ 4]^T$ and again performing 10 steps, results in the point $\boldsymbol{x}^{(10)} = [2.1753 \ \ 0.0068]^T$, which corresponds to the critical point $\boldsymbol{c}_3$ (a saddle point). The steps of this second calculation are shown in the following table:

$\boldsymbol{x}^{(0)}$	$\boldsymbol{x}^{(1)}$	$\boldsymbol{x}^{(2)}$	$\boldsymbol{x}^{(3)}$	$\boldsymbol{x}^{(4)}$	$\boldsymbol{x}^{(5)}$
$\begin{bmatrix} -2.5000 \\ 4.0000 \end{bmatrix}$	$\begin{bmatrix} 1.3684 \\ 2.3810 \end{bmatrix}$	$\begin{bmatrix} 3.1422 \\ 1.3661 \end{bmatrix}$	$\begin{bmatrix} 2.4434 \\ 0.7543 \end{bmatrix}$	$\begin{bmatrix} 2.2054 \\ 0.4024 \end{bmatrix}$	$\begin{bmatrix} 2.1758 \\ 0.2091 \end{bmatrix}$
$\boldsymbol{x}^{(6)}$	$\boldsymbol{x}^{(7)}$	$\boldsymbol{x}^{(8)}$	$\boldsymbol{x}^{(9)}$	$\boldsymbol{x}^{(10)}$	
$\begin{bmatrix} 2.1753 \\ 0.1068 \end{bmatrix}$	$\begin{bmatrix} 2.1753 \\ 0.0540 \end{bmatrix}$	$\begin{bmatrix} 2.1753 \\ 0.0272 \end{bmatrix}$	$\begin{bmatrix} 2.1753 \\ 0.0136 \end{bmatrix}$	$\begin{bmatrix} 2.1753 \\ 0.0068 \end{bmatrix}$	

As expected, Newton's method is attracted by the saddle points and gets stuck there, which makes the steepest descent method work better here. The steepest descent method even finds the global minimum, when starting at the second point $\boldsymbol{x}^{(0)} = [\text{-}2.5 \ \ 4]^T$.